Code No: 158DB **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, July/August - 2022
BASICS OF VIRTUAL INSTRUMENTATION
(Common to CSE, IT)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
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1.a) Explain the historical perspective and evolution of virtual instruments and draw the block diagram of a virtual instrument and describe the functions of each block.
b) Explain the programming languages used in development of VIs and describe the Lab view and its features. [8+7]

2.a) Explain the graphical programming in data flow and compare its operational performance with conventional data flow programming.
b) Compare and contrast the virtual instruments with traditional instruments. [8+7]

3. Explain the following programming techniques in VI
a) VIs and sub-VIs
b) Case and sequence structures
c) Instrument drivers. [5+5+5]

4. Describe the following programming methods for creating
a) FOR and WHILE loops
b) Graphs and Arrays
c) Formula nodes. [5+5+5]

5. Describe the following Interface standards with neat sketches
a) GPIB
b) USB 2.0
c) PCMCIA and VXI. [5+5+5]

6.a) Draw the block diagram of data acquisition hardware diagram and explain about Multiplexers, Sample and Hold and data converters used in it.
b) Explain in detail about ethernet control of PXI, VISA and IVI with neat sketches. [8+7]

7. Explain in detail about the following applications of Virtual instrumentation
a) Instrument control using RS232C and IEEE488
b) Development of Real-time systems and embedded systems. [8+7]

8.a) Illustrate the steps involved in the development of Lab VIEW-based motion control system with the flow chart and a neat schematic diagram.
b) Explain various VI tool sets used in development of VI based systems. [8+7]

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